

Input:

- V . Set of nodes
- E_s . Set of scheduled edges $(v, u, \text{dep}, \text{arr})$
- E_t . Set of transfer edges (v, w, Δ)
- $\text{country}(v)$. Country associated with node v
- $R(v, b)$. Precomputed reachable country sets
- $MTT(c)$. Precomputed minimum transit times for each country
- $\text{bucket}(t)$. Time bucket mapping function
- record . Current record to beat (e.g., 13). Static.
- rollout_freq . Number of iterations between rollouts (e.g., 1000)
- GlobalMinTime . Minimum realistic time to visit a country (e.g., 30 mins)

Output:

- P . Sequence of labels representing a route within 24 hours

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1: Initialize  $\text{Labels}(v) \leftarrow \emptyset$  for all  $v \in V$ 
2: Initialize priority queue  $Q \leftarrow \emptyset$  // ordered by decreasing score
3: Initialize best label  $\leftarrow \text{null}$ 
4:  $\text{iters} \leftarrow 0$  // Iteration counter for rollouts
5: // Initialize labels at departures
6: for all  $v \in V$  do
7:   for all scheduled edges  $(v, u, \text{dep}, \text{arr}) \in E_s$  do
8:      $L \leftarrow (u, \text{arr}, \{\text{country}(u)\}, \text{pred} = \text{null}, \text{start\_time} = \text{dep})$ 
9:      $\text{Labels}(u) \leftarrow \text{Labels}(u) \cup \{L\}$ 
10:    compute score( $L$ )
11:     $Q \leftarrow Q \cup \{L\}$ 
12: // Main loop
13: while  $Q \neq \emptyset$  do
14:   extract  $L = (v, t, C, \text{pred}, \text{start\_time})$  with maximum score from  $Q$ 
15:   // — GLOBAL HARD FLOOR PRUNING —
16:    $\text{needed} \leftarrow (\text{record} + 1) - |C|$ 
17:    $\text{remaining\_time} \leftarrow 1440 - (t - \text{start\_time})$ 
18:   // If we need countries but lack the physical time (e.g., j 30 mins per country), kill it.
19:   if  $\text{needed} > 0 \wedge \text{remaining\_time} < (\text{needed} \times \text{GlobalMinTime})$  then
20:     continue
21:   // —————
22:   // — MONTE CARLO ROLLOUT TRIGGER —
23:    $\text{iters} \leftarrow \text{iters} + 1$ 
24:   if  $\text{iters} \bmod \text{rollout\_freq} = 0$  then
25:     // Unconditional rollout for any label that survived Hard Floor
26:      $L_{\text{rollout}} \leftarrow \text{Algorithm 7}(L, E_s, E_t, \text{country})$ 
27:     if  $\text{best label} = \text{null} \vee |L_{\text{rollout}}.C| > |\text{best label}.C|$  then
28:        $\text{best label} \leftarrow L_{\text{rollout}}$ 
29:       // Early stopping if rollout beats the static record
30:       if  $|\text{best label}.C| > \text{record}$  then
31:         break
32:   // —————

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33: // Enforce 24-hour journey limit (1440 absolute minutes)
34: if remaining_time < 0 then
35:   continue
36: // Upper bound computation (Time-Budgeted)
37:  $b \leftarrow \text{bucket}(t)$ 
38:  $U \leftarrow R(v, b) \setminus C$ 
39: Sort  $U$  as a sequence  $(c_1, c_2, \dots, c_k)$  such that  $MTT(c_1) \leq MTT(c_2) \leq \dots \leq MTT(c_k)$ 
40: time_spent  $\leftarrow$  0
41: UB  $\leftarrow$  0
42: for  $i = 1$  to  $|U|$  do
43:   time_spent  $\leftarrow$  time_spent +  $MTT(c_i)$ 
44:   if time_spent  $\leq$  remaining_time then
45:     UB  $\leftarrow$  UB + 1
46:   else
47:     break
48: // Record-based pruning (Using static record)
49: if  $|C| + \text{UB} \leq \text{record}$  then
50:   continue
51: // Update best label from normal queue expansion
52: if best_label = null  $\vee |C| > |\text{best\_label}.C|$  then
53:   best_label  $\leftarrow$   $L$ 
54: // Early stopping if normal expansion beats the static record
55: if  $|C| > \text{record}$  then
56:   break
57: // Extend via scheduled edges
58: for all  $(v, u, \text{dep}, \text{arr}) \in E_s$  do
59:   if  $\text{dep} \geq t$  &  $\text{dep} - \text{start\_time} \leq 1440$  then
60:      $t_{\text{new}} \leftarrow \text{arr}$ 
61:     if  $t_{\text{new}} - \text{start\_time} > 1440$  then
62:       continue
63:      $C' \leftarrow C \cup \{\text{country}(u)\}$ 
64:      $L' \leftarrow (u, t_{\text{new}}, C', \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
65:     if  $\nexists L'' \in \text{Labels}(u)$  such that
66:        $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C' \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C'))$  then
67:         remove all  $L'' \in \text{Labels}(u)$  such that
68:            $(t_{\text{new}} \leq L''.t \wedge C' \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C' \neq L''.C))$ 
69:          $\text{Labels}(u) \leftarrow \text{Labels}(u) \cup \{L'\}$ 
70:         compute score( $L'$ )
71:          $Q \leftarrow Q \cup \{L'\}$ 
72: // Extend via transfers
73: for all  $(v, w, \Delta) \in E_t$  do
74:    $t_{\text{new}} \leftarrow t + \Delta$ 
75:   if  $t_{\text{new}} - \text{start\_time} > 1440$  then
76:     continue
77:    $L' \leftarrow (w, t_{\text{new}}, C, \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
78:   if  $\nexists L'' \in \text{Labels}(w)$  such that
79:      $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C))$  then
80:       remove all  $L'' \in \text{Labels}(w)$  such that
81:          $(t_{\text{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C \neq L''.C))$ 
82:        $\text{Labels}(w) \leftarrow \text{Labels}(w) \cup \{L'\}$ 

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79:   |   |   |   compute score( $L'$ )
80:   |   |   |    $Q \leftarrow Q \cup \{L'\}$ 
81: // Reconstruct path
82:  $P \leftarrow$  empty sequence
83:  $L \leftarrow$  best label
84: while  $L \neq$  null do
85:   |   prepend  $L$  to  $P$ 
86:   |    $L \leftarrow L.\text{pred}$ 
87: return  $P$ 

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After running Algorithm 8 and debugging, we observed an important issue: most of labels that were object of the rollouts were very close to the time-line end, with a very low country count. For example: [ROLLOUT] iters=316,000 — pre t=1373 — pre —C—=7 — post —C—=7.

This led to the idea of adding a global hard floor pruning mechanism. This uses a minimum realistic time per country (set arbitrarily and, of course, optimistically to avoid missing complicated but feasible routes) to aggressively prune labels that cannot possibly reach the record even with optimal remaining paths. Additionally, the rollout trigger is optimized to only execute when there is a comfortable time buffer, and the record is dynamically updated during search to enable immediate pruning of the queue.

Hence, this is a very similar algorithm to Algorithm 8, but with the addition of the global hard floor pruning mechanism which we hope can drastically reduce the labels that stay in the queue and are expanded, hoping for a faster convergence.